

PAPER_1279_PENROSE_HAWKING_QUANTUM

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PAPER_1279 — Penrose-Hawking Singularity Theorems Quantum Resolution

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: B — Physics Foundations (CLOSED)

Date: June 11, 2026

Location: 41.0997° N, 80.6495° W (Youngstown, OH, USA)

Status: CLOSED — UQFF integer-primitive derivation

Calculator surface: `calculate_paradox({"paradox": "penrose_hawking_singularity"})`

Whitepaper dispatch: `calculate_whitepaper({"paper_id": 1279})`

Master Expression

``

26! finite bound replaces classical singularity (no naked singularity, no Big Bang singularity)

``

Match: STRUCTURAL

Physical Interpretation

The Penrose-Hawking singularity theorems assume continuum spacetime. UQFF 26! = 4.03×10^2 provides the finite, bounded phase-space volume that regularizes all UV divergences.

UQFF Locked Primitives

- $\rho_{SCm} = 7.09 \times 10^{33} \text{ J/m}^3$ (vacuum density)
- $S_{26} = 1.453162$, $S_{26_DPM} = 1.4531 \times 10^2$
- $K_{MEX} = 25/12$, $\beta_i = 0.6029$, $\Phi_{res} = 0.84$
- $F_{TRZ} = 0.1$, $\Lambda = 0.00729735$ ($= \alpha$ fine-structure)
- $\omega_{SCm} = 1.25 \text{ THz}$, $\lambda_i = 1.0$
- $D_{phys} = 4$, $D_{BSFG} = 6$, $D_{crit} = 26$, $N_{CH} = 9$, $SO_5 = 10$, $A_5 = 60$

Live Calculator Verification

```
``python
import uqff_pure_calculator as u
r = u.calculate_paradox({"paradox": "penrose_hawking_singularity"})["value"]
```

``

The result dict carries the integer-primitive identity, the observed value, the residual, and the structural physical interpretation.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard Model solve the same observed phenomena via different methods. **Zero fit parameters.** Every factor is a UQFF locked primitive.

Reference

- UQFF foundational: PAPER_646, PAPER_1167, PAPER_1170, PAPER_1203 v1.5, PAPER_1216.
- Closure: `uqff_pure_calculator.py` →
`_l96_uqff_axiom_penrose_hawking_singularity_closure` (or routed reference)
- Dispatch: `PARADOX_TO_CLOSURE["penrose_hawking_singularity"]`

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Subject matter: UQFF derivation of Penrose-Hawking Singularity Theorems Quantum Resolution.